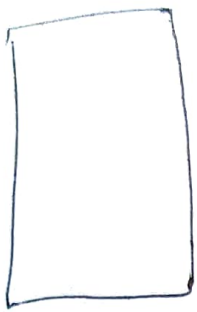


Q1)



4GB

1 word = 2 bytes

Max. Memory =

$$\underbrace{4 \times 1024}_{\text{MB}} \times \underbrace{1024}_{\text{KB}} \times \underbrace{1024}_{\text{bytes}}$$

$$= 2^2 \times 2^{10} \times 2^{10} \times 2^{10}$$

$$= 2^{32} \text{ bytes}$$

Size of a word = 2 bytes

$$\therefore \text{No. of words} = \frac{2^{32}}{2}$$
$$= 2^{31}$$

So we need 31 bits for the address bus of the processor.

Q) A processor can support a maximum memory of 4GB; where the memory is word addressable (a word consists of two bytes). The size of the address bus of the processor is at least — bits.

Q2) A 64-bit processor can support a maximum memory of 8 GB, where the memory is word-addressable (one word is of 64 bits). The size of the address bus of the processor is atleast _____ bits.

- a) 30 b) 31 c) 32 d) None



$$\text{max. Memory} = \underbrace{2^3}_{\text{MB}} \times \underbrace{1024}_{\text{KB}} \times \underbrace{1024}_{\text{bytes}} \times 1024$$

$$= 2^3 \times 2^{10} \times 2^{10} \times 2^{10}$$

$$= 2^{33} \text{ bytes}$$

$$\begin{aligned} \text{Size of a word} &= 64 \text{ bits} \\ &= 8 \text{ bytes} = 2^3 \text{ bytes} \end{aligned}$$

$$\begin{aligned} 8 \text{ bit} &= 1 \text{ byte} \\ 64 \text{ bit} &= \frac{64}{8} \end{aligned}$$

$$\begin{aligned} \therefore \text{No. of words} &= \frac{2^{33}}{2^3} \\ &= 2^{33-3} = \underline{\underline{2^{30}}} \end{aligned}$$

So we need 30 bits for the address bus of the processor